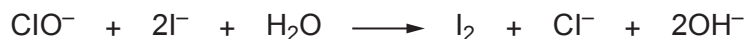
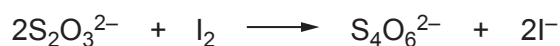


11. Chlorine can form a range of oxyanions with chlorine present in oxidation states including +1, +5 and +7.

- (a) Chlorate(I) ions,  $\text{ClO}^-$ , are commonly present in household bleaches. These ions can oxidise iodide ions to iodine.



Dewi took a  $25.0\text{ cm}^3$  sample of a household bleach and diluted it to form  $250\text{ cm}^3$  of solution in a standard volumetric flask. Four  $25.0\text{ cm}^3$  portions of the diluted solution were measured and placed in separate conical flasks and a spatula of solid potassium iodide was added to each. Each portion was titrated against standard sodium thiosulfate solution from a burette.



Dewi had access to five concentrations of sodium thiosulfate:  $2.00\text{ mol dm}^{-3}$ ,  $1.00\text{ mol dm}^{-3}$ ,  $0.500\text{ mol dm}^{-3}$ ,  $0.200\text{ mol dm}^{-3}$  and  $0.0500\text{ mol dm}^{-3}$ .

He selected the  $0.500\text{ mol dm}^{-3}$  solution and his results are shown in the table.

|                                                                      | 1    | 2    | 3    | 4    |
|----------------------------------------------------------------------|------|------|------|------|
| Volume of $\text{Na}_2\text{S}_2\text{O}_3$ solution / $\text{cm}^3$ | 6.45 | 6.40 | 6.50 | 6.45 |

- (i) Calculate the mean volume of sodium thiosulfate solution used.

[1]

Mean volume = .....  $\text{cm}^3$



- (ii) Find the concentration of sodium chlorate(I) in the initial bleach sample. [3]

Concentration = .....  $\text{mol dm}^{-3}$

- (iii) Bleach concentrations are often quoted on labels as % w/v, that is the mass of sodium chlorate(I) dissolved in  $100 \text{ cm}^3$  of water.

Calculate the % w/v concentration of sodium chlorate(I) in this bleach. [1]

Concentration = ..... % w/v

- (iv) The teacher tells Dewi that he has selected an inappropriate concentration of sodium thiosulfate solution.

Suggest which concentration of sodium thiosulfate he should have chosen.  
Give **two** reasons for your answer. [2]

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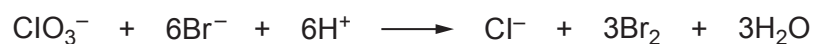
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- (b) Chlorate(V) anions,  $\text{ClO}_3^-$ , are strong oxidising agents and can oxidise bromide ions in acid solution to form bromine and chloride ions.



The initial rate of this reaction was measured at a temperature of 298 K using different concentrations of reactants.

| Experiment | Concentration of $\text{ClO}_3^-$ / $\text{mol dm}^{-3}$ | Concentration of $\text{Br}^-$ / $\text{mol dm}^{-3}$ | pH | Initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$ |
|------------|----------------------------------------------------------|-------------------------------------------------------|----|---------------------------------------------------|
| 1          | 0.30                                                     | 0.20                                                  | 0  | $3.06 \times 10^{-7}$                             |
| 2          | 0.60                                                     | 0.20                                                  | 0  | $6.12 \times 10^{-7}$                             |
| 3          | 0.30                                                     | 0.60                                                  | 0  | $9.18 \times 10^{-7}$                             |
| 4          | 0.30                                                     | 0.60                                                  | 1  |                                                   |

- (i) I. Find the orders of reaction with respect to chlorate(V) and bromide ions. [2]

You **must** show your working.

Order with respect to chlorate(V) ions .....

Order with respect to bromide ions .....



- II. The student finds that the rate of reaction is proportional to  $[H^+]^3$ . This is described as third order with respect to hydrogen ions.

Find the expected initial rate for experiment 4.

[2]

Initial rate = .....  $\text{mol dm}^{-3} \text{s}^{-1}$

- (ii) Chemists often use an approximate rule that the rate of a reaction doubles when the temperature is increased by  $10^\circ\text{C}$ .

Show that this rule is true if this reaction is warmed from 298 K to 308 K. The activation energy for the reaction is  $52.8 \text{ kJ mol}^{-1}$ .

[3]

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- (c) The acid formed from chlorate(VII) ions is commonly called perchloric acid,  $\text{HClO}_4$ . It is a very strong acid and is commonly classed as a superacid as it is a stronger acid than sulfuric acid.

- (i) State how the  $K_a$  value of a stronger acid will compare with that of a weaker acid, giving a reason. [1]

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- (ii) The concentration of  $25.0 \text{ cm}^3$  of an aqueous solution of perchloric acid can be found by adding an excess of calcium carbonate and then measuring the amount of carbon dioxide gas released.

This can be done by either:

- measuring the volume of the gas released using a gas syringe, or
- measuring the mass lost as the carbon dioxide is released by placing the reaction flask on a digital balance.

Joe uses the first method to study his sample of perchloric acid and finds that the reaction releases  $87 \text{ cm}^3$  of carbon dioxide at  $30^\circ\text{C}$  and 1 atm pressure.

Heledd uses the second method and finds that the mixture loses a total of 0.1533 g during the reaction.



- I. Find the number of moles of carbon dioxide released in each experiment and hence state whether the two solutions provided to Joe and Heledd are of the same concentration. [4]

You **must** show your working and give your answers to **appropriate** numbers of significant figures.

Moles of CO<sub>2</sub> using Joe's method = ..... mol

Moles of CO<sub>2</sub> using Heledd's method = ..... mol

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- II. Suggest, from the data provided, which of the two methods will give better results. Give a reason for your suggestion. [1]

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- III. Isha suggests using excess magnesium in place of calcium carbonate for Joe and Heledd's experiments. This will produce hydrogen gas in place of carbon dioxide. Explain which of the two methods will suffer the greater loss in accuracy. [2]

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(iii) One common salt of perchloric acid is ammonium perchlorate.

Suggest a pH value for a solution of ammonium perchlorate. Give a reason for your answer. [2]

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**END OF PAPER**

Examiner  
only

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